

CSL351 : COMPUTER NETWORKS

Assignment - 4 (Application Layer Protocols)

Name - Ashish Ranjan
ID no. - 12340370

Pcap File Link :

<https://drive.google.com/file/d/1HznmlxKHPmFK3BU6gBT8ON1Di4Lqpx0t/view?usp=sharing>

PART 1 (HTTP)

QUES 1.

COMMAND USED : `http.request.method=="GET" &&
ip.addr==192.168.10.115`

There are total of 36 http packets including both the request to the iitbhilai server and the response from the iitbhilaisherver.

Statistics			
Measurement	Captured	Displayed	Marked
Packets	7824	36 (0.5%)	—
Time span, s	83.716	6.743	—
Average pps	93.5	5.3	—
Average packet size, B	2778	779	—
Bytes	21734943	28053 (0.1%)	0
Average bytes/s	259 k	4,160	—
Average bits/s	2,077 k	33 k	—

For embedded content :

COMMAND USED : `ip.addr==192.168.10.115 && (http.accept contains
"png" or http.accept contains "jpeg" or http.accept contains "webp")`

Measurement	Captured	Displayed	Marked
Packets	7824	24 (0.3%)	—
Time span, s	83.716	6.743	—
Average pps	93.5	3.6	—
Average packet size, B	2778	797	—
Bytes	21734943	19124 (0.1%)	0
Average bytes/s	259 k	2,835	—
Average bits/s	2,077 k	22 k	—

ip.addr==192.168.10.115 && (http.accept contains "png" or http.accept contains "jpeg" or http.accept contains "webp")						
Packet list	Narrow & Wide		Case sensitive	String		
No.	Time	Source	Destination	Protocol	Length	Info
1096	11.896485	10.50.33.117	192.168.10.115	HTTP	801	GET / HTTP/1.1
1417	13.072850	10.50.33.117	192.168.10.115	HTTP	790	GET /index.php?pid=img_logo HTTP/1.1
1429	13.098280	10.50.33.117	192.168.10.115	HTTP	793	GET /index.php?pid=img_ppc2026 HTTP/1.1
1504	13.632772	10.50.33.117	192.168.10.115	HTTP	799	GET /index.php?pid=fifth_convocation HTTP/1.1
1620	14.065163	10.50.33.117	192.168.10.115	HTTP	803	GET /index.php?pid=independence_day_79th HTTP/1.1
1621	14.065452	10.50.33.117	192.168.10.115	HTTP	791	GET /index.php?pid=yoga_2025 HTTP/1.1
1653	14.124671	10.50.33.117	192.168.10.115	HTTP	787	GET /index.php?pid=img_transparent HTTP/1.1
2341	15.524085	10.50.33.117	192.168.10.115	HTTP	805	GET /index.php?pid=WorldEnvironmentDay2025 HTTP/1.1
2346	15.573286	10.50.33.117	192.168.10.115	HTTP	809	GET /index.php?pid=archaeology_archives_museum HTTP/1.1
2365	15.628042	10.50.33.117	192.168.10.115	HTTP	797	GET /index.php?pid=img_republicday HTTP/1.1
2674	16.014915	10.50.33.117	192.168.10.115	HTTP	793	GET /index.php?pid=cclt_launch HTTP/1.1
3063	16.671776	10.50.33.117	192.168.10.115	HTTP	816	GET /index.php?pid=Technology_Prioritization_Workshop HTTP/1.1
3222	16.996491	10.50.33.117	192.168.10.115	HTTP	792	GET /index.php?pid=meraz_24_1 HTTP/1.1
3404	17.128387	10.50.33.117	192.168.10.115	HTTP	799	GET /index.php?pid=independence_2024 HTTP/1.1
3593	17.321027	10.50.33.117	192.168.10.115	HTTP	792	GET /index.php?pid=gi_tagging HTTP/1.1
3597	17.329848	10.50.33.117	192.168.10.115	HTTP	806	GET /index.php?pid=indo_german_workshop2023 HTTP/1.1
3694	17.414194	10.50.33.117	192.168.10.115	HTTP	787	GET /index.php?pid=news3 HTTP/1.1
3819	17.542455	10.50.33.117	192.168.10.115	HTTP	787	GET /index.php?pid=news1 HTTP/1.1
3880	17.606286	10.50.33.117	192.168.10.115	HTTP	787	GET /index.php?pid=news2 HTTP/1.1
3960	17.778185	10.50.33.117	192.168.10.115	HTTP	796	GET /index.php?pid=vigilance_2023 HTTP/1.1
3999	17.817605	10.50.33.117	192.168.10.115	HTTP	795	GET /index.php?pid=img_top_rajiv HTTP/1.1
4133	18.011288	10.50.33.117	192.168.10.115	HTTP	797	GET /index.php?pid=img_glimpses_pc HTTP/1.1
4590	18.464782	10.50.33.117	192.168.10.115	HTTP	804	GET /index.php?pid=swakshata_hi_seva_2023 HTTP/1.1
4654	18.639974	10.50.33.117	192.168.10.115	HTTP	798	GET /index.php?pid=teachersday_2023 HTTP/1.1

There are total 24 embedded content packets

For text content :

COMMAND USED : ip.addr==192.168.10.115 && (http.accept and !(http.accept contains "image") and !(http.accept contains "video"))

Measurement	Captured	Displayed	Marked
Packets	7824	12 (0.2%)	—
Time span, s	83.716	3.026	—
Average pps	93.5	4.0	—
Average packet size, B	2778	744	—
Bytes	21734943	8929 (0.0%)	0
Average bytes/s	259 k	2,950	—
Average bits/s	2,077 k	23 k	—

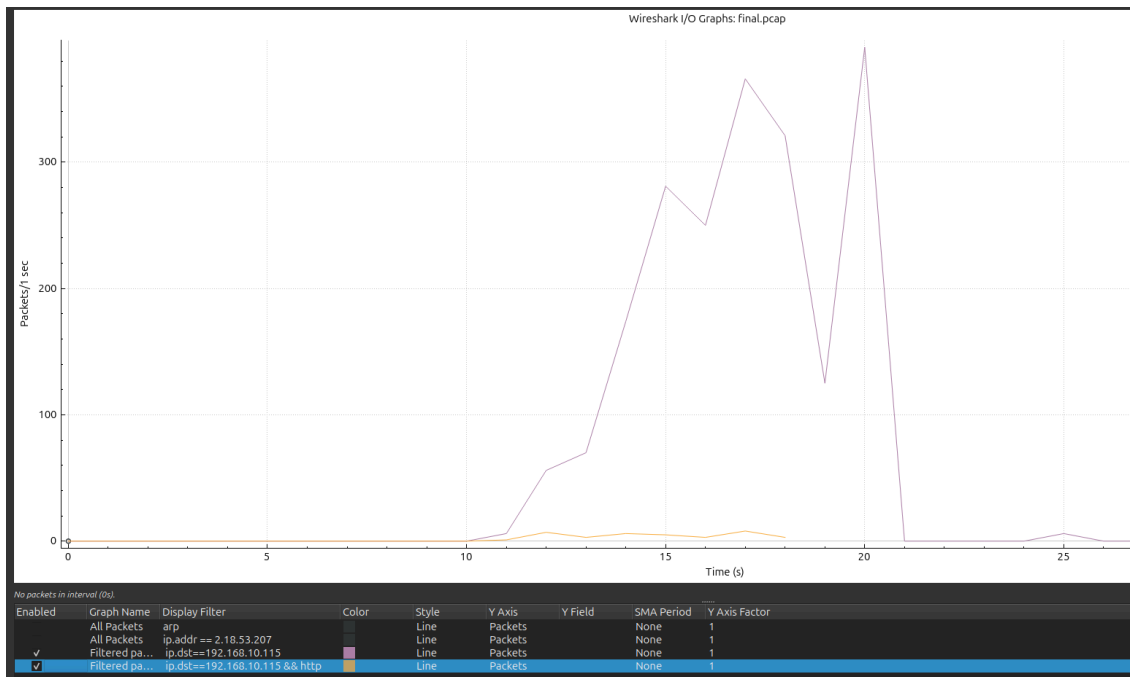
ip.addr==192.168.10.115 && (http.accept and !(http.accept contains "image") and !(http.accept contains "video"))						
Packet list	Narrow & Wide		Case sensitive	String		
No.	Time	Source	Destination	Protocol	Length	Info
1294	12.455056	10.50.33.117	192.168.10.115	HTTP	752	GET /index.php?pid=css_bootstrapmin HTTP/1.1
1318	12.567024	10.50.33.117	192.168.10.115	HTTP	745	GET /index.php?pid=css_style HTTP/1.1
1324	12.568529	10.50.33.117	192.168.10.115	HTTP	756	GET /index.php?pid=css_bootstrap_select HTTP/1.1
1374	12.812242	10.50.33.117	192.168.10.115	HTTP	754	GET /index.php?pid=css_fontawesomemin HTTP/1.1
1377	12.846045	10.50.33.117	192.168.10.115	HTTP	731	GET /index.php?pid=js_search HTTP/1.1
1390	12.901814	10.50.33.117	192.168.10.115	HTTP	734	GET /index.php?pid=js_jquerymin HTTP/1.1
1397	12.978725	10.50.33.117	192.168.10.115	HTTP	741	GET /index.php?pid=js_bootstrap_select HTTP/1.1
1623	14.065948	10.50.33.117	192.168.10.115	HTTP	737	GET /index.php?pid=js_bootstrapmin HTTP/1.1
1658	14.136675	10.50.33.117	192.168.10.115	HTTP	738	GET /index.php?pid=js_effi_cryptojs HTTP/1.1
1753	14.493870	10.50.33.117	192.168.10.115	HTTP	749	GET /index.php?pid=js_effi_cryptojs_hmac HTTP/1.1
2302	15.454365	10.50.33.117	192.168.10.115	HTTP	748	GET /index.php?pid=js_effi_cryptojs_enc HTTP/1.1
2323	15.481109	10.50.33.117	192.168.10.115	HTTP	744	GET /index.php?pid=js_effi_serviceutil HTTP/1.1

There are total 12 text packets.

I/O Graphs :

pink : all the packets sent to iitbhillai

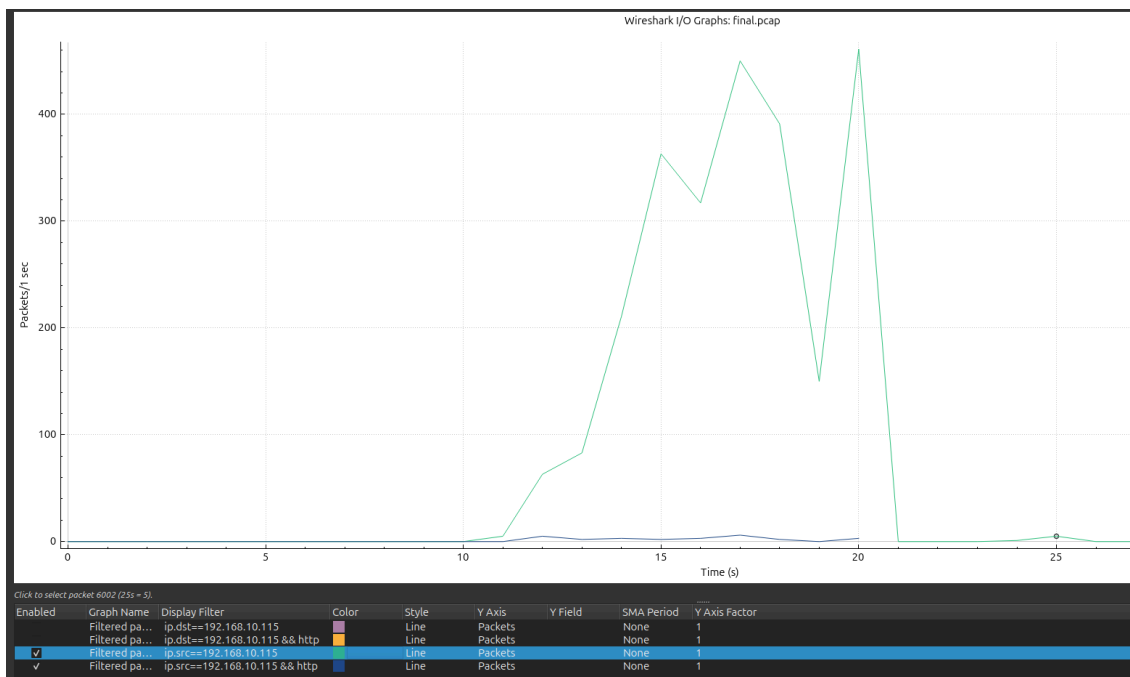
yellow : http packets sent to iitbhillai



Packets received from iitbhillai

green : all packets received from iitbhillai

blue : http packets received from iitbhillai



QUES 2.

COMMAND USED : ip.src==192.168.10.115 && http

Statistics			
Measurement	Captured	Displayed	Marked
Packets	7824	26 (0.3%)	1 (0.0%)
Time span, s	83.716	8.580	—
Average pps	93.5	3.0	—
Average packet size, B	2778	6038	801
Bytes	21734943	156982 (0.7%)	801 (0.0%)
Average bytes/s	259 k	18 k	—
Average bits/s	2,077 k	146 k	—

Total of 156982 bytes of data was received in response to the get requests.

QUES 3.

COMMAND USED : ip.addr==192.168.10.115 && http.response.code!=200

There were 0 packets with response other than 200 .

ip.addr==192.168.10.115 && http.response.code!=200

Packet list

Narrow & Wide

☐ Case sensitive

String

No.	Time	Source	Destination	Protocol	Length	Info
-----	------	--------	-------------	----------	--------	------

Measurement	Captured	Displayed	Marked
Packets	7824	—	1 (0.0%)
Time span, s	83.716	—	—
Average pps	93.5	—	—
Average packet size, B	2778	—	801
Bytes	21734943	0	801 (0.0%)
Average bytes/s	259 k	—	—
Average bits/s	2,077 k	—	—

QUES 4.

Since under a single stream there are multiple connections which shows these are persistent connections, which alternatively verified with the `http.connection=="Keep-Alive"`

Total persistent http packets count is 26

Elapsed:	00:01:23		
Capture			
Hardware:	Unknown		
OS:	Unknown		
Application:	Unknown		
Interfaces			
Interface	Dropped packets	Capture filter	Link type
Unknown	Unknown	Unknown	Ethernet
			Packet size limit (snaplen)
			262144 bytes
Statistics			
Measurement	Captured	Displayed	Marked
Packets	7824	26 (0.3%)	—
Time span, s	83.716	8.580	—
Average pps	93.5	3.0	—
Average packet size, B	2778	6038	—
Bytes	21734943	156982 (0.7%)	0
Average bytes/s	259 k	18 k	—
Average bits/s	2,077 k	146 k	—

http.connection contains "Keep-Alive"									
Packet list									
No.	Time	Source	Destination	Protocol	Length	Info			
1182	12.113187	192.168.10.115	10.50.33.117	TLSv1.2	3549	HTTP/1.1 200 OK (text/html)			
1372	12.889797	192.168.10.115	10.50.33.117	HTTP	6636	HTTP/1.1 200 OK (text/css)			
1376	12.844869	192.168.10.115	10.50.33.117	HTTP	1722	HTTP/1.1 200 OK (text/css)			
1388	12.897109	192.168.10.115	10.50.33.117	HTTP	1255	HTTP/1.1 200 OK (text/css)			
1395	12.973729	192.168.10.115	10.50.33.117	HTTP	947	HTTP/1.1 200 OK (text/javascript)			
1413	13.078937	192.168.10.115	10.50.33.117	HTTP	6847	HTTP/1.1 200 OK (text/css)			
1502	13.628963	192.168.10.115	10.50.33.117	HTTP	2297	HTTP/1.1 200 OK (PNG)			
1610	14.049796	192.168.10.115	10.50.33.117	HTTP	2591	HTTP/1.1 200 OK (text/javascript)			
1652	14.124293	192.168.10.115	10.50.33.117	HTTP	690	HTTP/1.1 200 OK (JPEG JFIF image)			
1751	14.491686	192.168.10.115	10.50.33.117	HTTP	1498	HTTP/1.1 200 OK (JPEG JFIF image)			
2339	15.518840	192.168.10.115	10.50.33.117	HTTP	879	HTTP/1.1 200 OK (text/javascript)			
2363	15.625856	192.168.10.115	10.50.33.117	HTTP	1536	HTTP/1.1 200 OK (text/javascript)			
2673	16.012705	192.168.10.115	10.50.33.117	HTTP	237	HTTP/1.1 200 OK (PNG)			
3061	16.665383	192.168.10.115	10.50.33.117	TLSv1.2	11029	HTTP/1.1 200 OK (image/jpeg)			
3220	16.994472	192.168.10.115	10.50.33.117	TLSv1.2	20100	HTTP/1.1 200 OK (JPEG JFIF image)			
3402	17.126679	192.168.10.115	10.50.33.117	TLSv1.2	19821	HTTP/1.1 200 OK (JPEG JFIF image)			
3692	17.413861	192.168.10.115	10.50.33.117	HTTP	1498	HTTP/1.1 200 OK (JPEG JFIF image)			
3816	17.541803	192.168.10.115	10.50.33.117	HTTP	14796	HTTP/1.1 200 OK (JPEG JFIF image)			
3879	17.695829	192.168.10.115	10.50.33.117	HTTP	1160	HTTP/1.1 200 OK (JPEG JFIF image)			
3958	17.777472	192.168.10.115	10.50.33.117	HTTP	1365	HTTP/1.1 200 OK (JPEG JFIF image)			
3997	17.817837	192.168.10.115	10.50.33.117	TLSv1.2	11342	HTTP/1.1 200 OK (JPEG JFIF image)			
4130	18.010018	192.168.10.115	10.50.33.117	TLSv1.2	12234	HTTP/1.1 200 OK (JPEG JFIF image)			
4649	18.037108	192.168.10.115	10.50.33.117	HTTP	5092	HTTP/1.1 200 OK (PNG)			
5815	20.508614	192.168.10.115	10.50.33.117	HTTP	4134	HTTP/1.1 200 OK (PNG)			
5870	20.583496	192.168.10.115	10.50.33.117	HTTP	11212	HTTP/1.1 200 OK (PNG)			
5956	20.693374	192.168.10.115	10.50.33.117	TLSv1.2	12522	HTTP/1.1 200 OK (JPEG JFIF image)			

QUES 5.

Input:

58fcdf35beea57bca75391de358e8438fdefe9de14a4e97f8d7573fe67 added4df6fb9d7af
cff0e96f1affd9

Convert From:
Hexadecimal

Convert To:
PNG

Image Preview



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We have visited netflix.com for this question

DNS resolution :

First the computer asks the DNS ip of netflix.com in whose response it gets dradis.netflix.com , after that it gets

www.us-west-2.internal.dradis.netflix.com after which it gets apiproxy-website-nlb-prod-2-e98cb8cf33ff3581.elb.us-west-2.amazonaws.com which was resolved with the following IPs 44.234.232.238, 44.237.234.25, 44.242.60.85 .

There can be other DNS resolutions when needed by the website to load contents like the web.prod.cloud.netflix.com be other DNS resolutions when needed by the website to load contents like the web.prod.cloud.netflix.com

No.	Time	Source	Destination	Protocol	Length	Info
	6286.36.488771	10.50.33.117	192.168.10.87	DNS	86	Standard query 0x7e81 A www.netflix.com OPT
	6291.36.573946	192.168.10.87	10.50.33.117	DNS	459	Standard query response 0x7e81 A www.netflix.com CNAME www.dradis.netflix.com CNAME www.us-west

```

Wireshark · Packet 6291 · Final.pcap

Class: IN (0x0001)
Time to live: 16 (16 seconds)
Data length: 71
CNAME: apiproxy-website-nlb-prod-2-e98cb8cf33ff3581.elb.us-west-2.amazonaws.com
  ▾ apiproxy-website-nlb-prod-2-e98cb8cf33ff3581.elb.us-west-2.amazonaws.com: type A, class IN, addr 44.234.232.238
    Name: apiproxy-website-nlb-prod-2-e98cb8cf33ff3581.elb.us-west-2.amazonaws.com
    Type: A (1) (Host Address)
    Class: IN (0x0001)
    Time to live: 60 (1 minute)
    Data length: 4
    Address: 44.234.232.238
  ▾ apiproxy-website-nlb-prod-2-e98cb8cf33ff3581.elb.us-west-2.amazonaws.com: type A, class IN, addr 44.237.234.25
    Name: apiproxy-website-nlb-prod-2-e98cb8cf33ff3581.elb.us-west-2.amazonaws.com
    Type: A (1) (Host Address)
    Class: IN (0x0001)
    Time to live: 60 (1 minute)
    Data length: 4
    Address: 44.237.234.25
  ▾ apiproxy-website-nlb-prod-2-e98cb8cf33ff3581.elb.us-west-2.amazonaws.com: type A, class IN, addr 44.242.60.85
    Name: apiproxy-website-nlb-prod-2-e98cb8cf33ff3581.elb.us-west-2.amazonaws.com
    Type: A (1) (Host Address)
    Class: IN (0x0001)
    Time to live: 60 (1 minute)
    Data length: 4
0010 01 bd 99 df 00 00 3e 11 ea aa c0 a8 0a 57 0a 32  ....>....W.2

```

Three way handshake :

No.	Time	Source	Destination	Protocol	Length	Info
6294	36.587711	10.50.33.117	44.234.232.238	TCP	74	60928 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=730499948 TSecr=0 WS=128
6322	36.906029	44.234.232.238	10.50.33.117	TCP	74	443 → 60928 [SYN, ACK] Seq=0 Ack=1 Win=41352 Len=0 MSS=1296 SACK_PERM TSval=4177608203 TSecr=730499948 WS=128
6323	36.900709	10.50.33.117	44.234.232.238	TCP	66	60928 → 443 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=7305008261 TSecr=4177608203
6324	36.901558	10.50.33.117	44.234.232.238	TLSPv1.3	1887	Client Hello [SNI=www.netflix.com]
6352	37.210568	44.234.232.238	10.50.33.117	TCP	64	443 → 60928 [ACK] Seq=1 Ack=1822 Win=39936 Len=0 TSval=4177608514 TSecr=7305008262
6353	37.211569	44.234.232.238	10.50.33.117	TLSPv1.3	1384	Server Hello, Change Cipher Spec
6354	37.211536	44.234.232.238	10.50.33.117	TLSPv1.3	66	60928 → 443 [ACK] Seq=1822 Ack=1239 Win=63184 Len=0 TSval=7305008572 TSecr=4177608515
6357	37.214452	10.50.33.117	44.234.232.238	TLSPv1.3	2818	Encrypted Extensions, Certificate, Certificate Verify, Finished
6358	37.214465	10.50.33.117	44.234.232.238	TCP	66	60928 → 443 [ACK] Seq=1822 Ack=3991 Win=60416 Len=0 TSval=7305008575 TSecr=4177608515
6365	37.218797	10.50.33.117	44.234.232.238	TLSPv1.3	130	Change Cipher Spec, Finished

As we can see in this TCP stream the three way handshake happens between the our machine and the server netflix.com
Later after the handshake,

TLS Handshake :

After the Three-way handshake TLS handshake occurs which encrypts the data so its not read by others on the flow

6342	37.177851	142.251.222.110	10.50.33.117	QUIC	82 Initial, SCID=e9398e463f29394d, PKN: 1, ACK
6343	37.179345	142.251.222.110	10.50.33.117	QUIC	1292 Initial, SCID=e9398e463f29394d, PKN: 2, ACK, PADDING
6344	37.179346	142.251.222.110	10.50.33.117	QUIC	1292 Initial, SCID=e9398e463f29394d, PKN: 3, CRYPTO, PADDING
6345	37.179346	142.251.222.110	10.50.33.117	QUIC	1292 Initial, SCID=e9398e463f29394d, PKN: 4, CRYPTO, PADDING
6346	37.179722	10.50.33.117	142.251.222.110	QUIC	1292 Initial, DCID=e9398e463f29394d, PKN: 3, ACK, PADDING
6347	37.180985	142.251.222.110	10.50.33.117	QUIC	1292 Handshake, SCID=e9398e463f29394d, PKN: 5, CRYPTO
6348	37.180985	142.251.222.110	10.50.33.117	QUIC	1292 Handshake, SCID=e9398e463f29394d, PKN: 6, CRYPTO
6349	37.180985	142.251.222.110	10.50.33.117	QUIC	1292 Handshake, SCID=e9398e463f29394d, PKN: 7, CRYPTO
6350	37.181155	10.50.33.117	142.251.222.110	QUIC	81 Handshake, DCID=e9398e463f29394d, PKN: 4, ACK
6351	37.182324	10.50.33.117	142.251.222.110	QUIC	82 Handshake, DCID=e9398e463f29394d, PKN: 5, ACK
6352	37.210568	44.234.232.238	10.50.33.117	TCP	66 443 -> 60928 [ACK] Seq=1 Ack=1822 Win=39936 Len=0 TSval=4177608514 TSecr=730500262
6353	37.211509	44.234.232.238	10.50.33.117	TLSv1.3	1304 Server Hello, Change Cipher Spec
6354	37.211536	10.50.33.117	44.234.232.238	TCP	66 60928 -> 443 [ACK] Seq=1822 Ack=1239 Win=63104 Len=0 TSval=730500572 TSecr=4177608515
6355	37.211560	10.50.33.117	142.251.222.110	QUIC	79 Handshake, DCID=e9398e463f29394d, PKN: 7, PADDING, PING
6356	37.214441	142.251.222.110	10.50.33.117	HTTP3	868 Protected Payload (KP0), PKN: 9, STREAM(3), SETTINGS
6357	37.214452	44.234.232.238	10.50.33.117	TLSv1.3	2818 Encrypted Extensions, Certificate, Certificate Verify, Finished
6358	37.214465	10.50.33.117	44.234.232.238	TCP	66 60928 -> 443 [ACK] Seq=1822 Ack=3991 Win=60416 Len=0 TSval=730500575 TSecr=4177608515
6359	37.216240	10.50.33.117	142.251.222.110	QUIC	82 Handshake, DCID=e9398e463f29394d, PKN: 8, ACK
6360	37.217397	10.50.33.117	142.251.222.110	QUIC	125 Handshake, DCID=e9398e463f29394d, PKN: 9, CRYPTO
6361	37.217568	10.50.33.117	142.251.222.110	HTTP3	113 Protected Payload (KP0), DCID=e9398e463f29394d, PKN: 10, STREAM(2), SETTINGS
6362	37.218395	10.50.33.117	142.251.222.110	HTTP3	1288 Protected Payload (KP0), DCID=e9398e463f29394d, PKN: 11, ACK, STREAM(2), PRIORITY_UPDATE, STREAM_UPDATE
6363	37.218452	10.50.33.117	142.251.222.110	HTTP3	1292 [Illegal Fragments]
6364	37.218462	10.50.33.117	142.251.222.110	HTTP3	150 Protected Payload (KP0), DCID=e9398e463f29394d, PKN: 13, STREAM(10), STREAM(0), HEADERS
6365	37.218797	10.50.33.117	44.234.232.238	TLSv1.3	130 Change Cipher Spec, Finished

HTTP Requests :

After this handshake the actual content of the website loads including the html page, various css, images as we can see in the packet 6367 the first GET request. After reading the HTML the browser now makes use of the assets which was resolved earlier.

Time required for the webpage to load :

6284	36.474791	10.50.33.117	216.239.38.223	HTTP3	191 Protected Payload (KP0), DCID=f4cbe8d1e6f79381, PKN: 11, STREAM(2)
6285	36.480682	10.50.33.117	192.168.10.87	DNS	86 Standard query 0xb29d HTTPS www.netflix.com OPT
6286	36.480771	10.50.33.117	192.168.10.87	DNS	86 Standard query 0x7e81 A www.netflix.com OPT
6287	36.491786	fe80::396d:b00c:d00d:ff02::1		ICMPv6	86 Neighbor Advertisement fe80::396d:b00c:d00d:8ee8 (ovr) is at 4a:5b:40:20:2e002e

And the first DNS lookup happened at 36.480 at packet 6285.

6645	38.626920	44.234.232.238	10.50.33.117	HTTP2	2542 DATA[1]
6646	38.626943	10.50.33.117	44.234.232.238	TCP	66 60928 -> 443 [ACK] Seq=3354 Ack=83882 Win=122368 Len=0 TSval=730501
6647	38.630939	44.234.232.238	10.50.33.117	HTTP2	730 DATA[0], DATA[1] (text/html)
6648	38.630959	10.50.33.117	44.234.232.238	TCP	66 60928 -> 443 [ACK] Seq=3354 Ack=84546 Win=124928 Len=0 TSval=730501
6649	38.630976	10.50.33.117	192.168.10.87	ICMPv6	74 Protected Payload (KP0), DATA=fe80::396d:b00c:d00d:8ee8 (ovr) is at 4a:5b:40:20:2e002e

According to this TCP stream packet 6647 the html page of the website was fully loaded at that point of time i.e 38.638

So total time to load the webpage was 2.158 seconds.

Total of 88 connections including (48 TCP + 40UDP) which includes the background process for the website, ads, CDNs as we might be connected to different servers parallelly.

frame.time_relative:36.16 && (http || http2 || http3)

Wireshark · Conversations · final.pcap

Conversation Settings

Name resolution

Absolute start time

✓ Limit to display filter

Copy

Follow Stream...

Graph...

Protocol

Bluetooth

BPV7

DCCP

✓ Ethernet

FC

FDDI

IEEE 802.11

IEEE 802.15.4

✓ IPv4

✓ IPv6

IPX

JXTA

LTP

MPTCP

NCP

openSAFETY

Filter list for specific type

Ethernet · 83	IPv4 · 24	IPv6 · 81	TCP · 48	UDP · 40						
Address A	Address B		Packets	Bytes	Total Packets	Percent Filtered	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A
00:28:f8:93:79:ed	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
00:41:0e:9e:9a:95	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
00:41:0e:f5:b5:62	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
02:fc:02:13:01:e3	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
04:ec:d8:c3:3e:9e	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
08:a5:c8:62:17:6b	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
10:f6:0a:b1:b9:49	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
1a:83:8c:c0:ed:c6	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
20:0b:74:48:7f:5c	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
20:74:54:58:05:43	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
22:7f:b6:6b:07:b6	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
24:b2:b9:21:0a:2d	8ce9:ee:4a:52:4a		2	172 bytes	2	100.00%	2	172 bytes	0	0 bytes
28:c5:d2:26:8d:3d	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
2c:a7:ef:9b:f6:3b	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
30:03:c8:ac:cf:9d	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
30:f6:f1:c1:e1:84	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
34:0a:33:30:59:85	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
34:6f:24:b1:f5:89	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
34:7d:f6:c3:99:57	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
38:de:ad:88:13:2d	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
3c:64:cf:8d:58:45	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
40:ae:30:3d:c9:df	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
44:fa:66:56:9e:ef	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
46:f2:78:5e:0a:d6	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
48:ef:1c:1d:1e:54	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
4a:09:47:19:76:78	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
4a:25:dc:ce:3d:6e	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
4a:5a:50:91:38:c3	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
4c:23:38:32:3d:5b	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
50:49:b0:aa:ac:c8	8ce9:ee:4a:52:4a		1	78 bytes	1	100.00%	1	78 bytes	0	0 bytes
52:17:82:19:09:8a	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
5a:08:19:36:2e:81	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
60:e9:aa:70:2b:b5	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
60:e9:aac:f2:cf	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes
60:ff:0a:1f:5b:00	8ce9:ee:4a:52:4a		1	86 bytes	1	100.00%	1	86 bytes	0	0 bytes

Help

Close

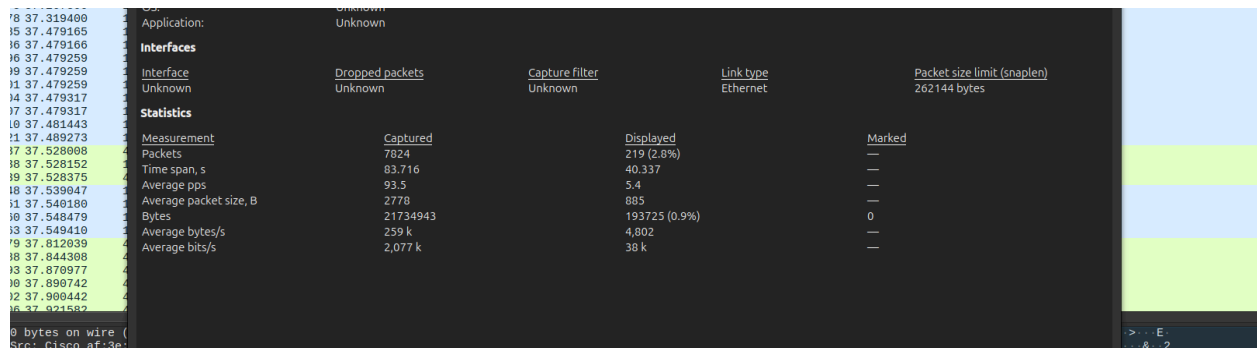
These connections are persistent.

No.	Time	Source	Destination	Protocol	Length	Info
6244	36.507711	10.50.33.117	44.234.232.238	TCP	74	60928 → 443 [SYN] Seq=0 Win=64249 Len=0 MSS=1460 SACK_PERM TSV=730499948 TSecr=0 WS=128
6322	36.900629	44.234.232.238	10.50.33.117	TCP	74	443 → 60928 [SYN, ACK] Seq=0 Ack=1 Win=41352 Len=0 MSS=1250 SACK_PERM TSV=4177608203 TSecr=730499948 WS=512
6323	36.900709	10.50.33.117	44.234.232.238	TCP	60	60928 → 443 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSV=730500261 TSecr=4177608203
6324	36.901558	10.50.33.117	44.234.232.238	TLSv1.3	1687	Client Hello (SHA256, net/tls.com)
6352	37.210568	44.234.232.238	10.50.33.117	TCP	60	443 → 60928 [ACK] Seq=1 Ack=1822 Win=39936 Len=0 TSV=4177608514 TSecr=730500262
6353	37.211509	44.234.232.238	10.50.33.117	TLSv1.3	1304	Server Hello, Change Cipher Spec
6354	37.211550	10.50.33.117	44.234.232.238	TCP	60	60928 → 443 [ACK] Seq=1822 Ack=3991 Win=64256 Len=0 TSV=730500572 TSecr=4177608515
6357	37.214452	44.234.232.238	10.50.33.117	TLSv1.3	2818	Encrypted Extensions, Certificate, Certificate Verify, Finished
6358	37.214465	10.50.33.117	44.234.232.238	TCP	60	60928 → 443 [ACK] Seq=1822 Ack=3991 Win=64256 Len=0 TSV=730500575 TSecr=4177608515
6365	37.216797	10.50.33.117	44.234.232.238	TLSv1.3	130	Change Cipher Spec, Finished
6366	37.219044	10.50.33.117	44.234.232.238	HTTP2	150	Magic, SETTINGS[0], WINDOW_UPDATE[0]
6367	37.219118	10.50.33.117	44.234.232.238	HTTP2	1411	HEADERS[1]: GET /in/
6436	37.527718	44.234.232.238	10.50.33.117	TCP	60	443 → 60928 [ACK] Seq=3991 Ack=3323 Win=38912 Len=0 TSV=4177608831 TSecr=730500579
6437	37.528088	44.234.232.238	10.50.33.117	HTTP2	550	SETTINGS[0], WINDOW_UPDATE[0]
6438	37.528152	10.50.33.117	44.234.232.238	HTTP2	97	SETTINGS[0]
6439	37.528375	44.234.232.238	10.50.33.117	HTTP2	97	SETTINGS[0]
6465	37.568443	10.50.33.117	44.234.232.238	TCP	60	60928 → 443 [ACK] Seq=3354 Ack=4508 Win=60032 Len=0 TSV=730500929 TSecr=4177608832
6477	37.811759	44.234.232.238	10.50.33.117	TCP	1384	443 → 60928 [ACK] Seq=4508 Ack=3323 Win=38912 Len=1238 TSV=4177609103 TSecr=730500579 [TCP segment of a reassembled P...
6478	37.811772	10.50.33.117	44.234.232.238	TCP	60	60928 → 443 [ACK] Seq=3354 Ack=5746 Win=58880 Len=0 TSV=730501172 TSecr=4177609103
6479	37.812039	44.234.232.238	10.50.33.117	HTTP2	323	HEADERS[1]: 200 OK
6480	37.812044	10.50.33.117	44.234.232.238	TCP	60	60928 → 443 [ACK] Seq=3354 Ack=6003 Win=58624 Len=0 TSV=730501172 TSecr=4177609103
6481	37.813044	44.234.232.238	10.50.33.117	TCP	2542	443 → 60928 [ACK] Seq=6003 Ack=3323 Win=38912 Len=2476 TSV=4177609104 TSecr=730500579 [TCP segment of a reassembled P...
6482	37.813054	10.50.33.117	44.234.232.238	TCP	60	60928 → 443 [ACK] Seq=3354 Ack=8478 Win=56192 Len=0 TSV=730501173 TSecr=4177609104
6483	37.829600	44.234.232.238	10.50.33.117	TCP	1384	443 → 60928 [ACK] Seq=8478 Ack=3323 Win=38912 Len=1238 TSV=4177609114 TSecr=730500579 [TCP segment of a reassembled P...
6484	37.829608	10.50.33.117	44.234.232.238	TCP	60	60928 → 443 [ACK] Seq=3354 Ack=9717 Win=55040 Len=0 TSV=730501190 TSecr=4177609114
6488	37.844308	44.234.232.238	10.50.33.117	HTTP2	1189	DATA[1]
6489	37.844336	10.50.33.117	44.234.232.238	TCP	60	60928 → 443 [ACK] Seq=3354 Ack=10840 Win=54016 Len=0 TSV=730501204 TSecr=4177609125
6491	37.859604	44.234.232.238	10.50.33.117	TCP	1384	443 → 60928 [ACK] Seq=10840 Ack=3323 Win=38912 Len=1238 TSV=4177609135 TSecr=730500579 [TCP segment of a reassembled F...
6492	37.859629	10.50.33.117	44.234.232.238	TCP	60	60928 → 443 [ACK] Seq=3354 Ack=12678 Win=52864 Len=0 TSV=730501220 TSecr=4177609135
6493	37.870977	44.234.232.238	10.50.33.117	HTTP2	1384	DATA[1]
6494	37.870996	10.50.33.117	44.234.232.238	TCP	60	60928 → 443 [ACK] Seq=3354 Ack=13316 Win=51712 Len=0 TSV=730501231 TSecr=4177609146
6495	37.881224	44.234.232.238	10.50.33.117	TCP	1384	443 → 60928 [ACK] Seq=13316 Ack=3323 Win=38912 Len=1238 TSV=4177609157 TSecr=730500579 [TCP segment of a reassembled F...
6496	37.881253	10.50.33.117	44.234.232.238	TCP	60	60928 → 443 [ACK] Seq=3354 Ack=14554 Win=50560 Len=0 TSV=730501241 TSecr=4177609157
6500	37.890742	44.234.232.238	10.50.33.117	HTTP2	1384	DATA[1]
6501	37.890761	10.50.33.117	44.234.232.238	TCP	60	60928 → 443 [ACK] Seq=3354 Ack=15792 Win=49408 Len=0 TSV=730501251 TSecr=4177609167
6502	37.890842	44.234.232.238	10.50.33.117	HTTP2	1384	DATA[1]
6503	37.890897	10.50.33.117	44.234.232.238	TCP	60	60928 → 443 [ACK] Seq=3354 Ack=17030 Win=48256 Len=0 TSV=730501261 TSecr=4177609178
6504	37.890927	44.234.232.238	10.50.33.117	TCP	1384	443 → 60928 [ACK] Seq=17030 Ack=3354 Win=38912 Len=1238 TSV=4177609189 TSecr=730500888 [TCP segment of a reassembled F...
6505	37.890944	10.50.33.117	44.234.232.238	TCP	60	60928 → 443 [ACK] Seq=3354 Ack=18268 Win=47104 Len=0 TSV=730501269 TSecr=4177609189
6506	37.921582	44.234.232.238	10.50.33.117	HTTP2	1384	DATA[1]
6507	37.921608	10.50.33.117	44.234.232.238	TCP	60	60928 → 443 [ACK] Seq=3354 Ack=19506 Win=45952 Len=0 TSV=730501282 TSecr=4177609200

As this whole data and exchange of info happened in one sequence and not broken frequently.

The different objects transferred in this connection include html, json files, image requested through API and also assets data which is not explicitly stated.

This accumulates to a total count of 219 objects which includes fragmented objects too.



The image shows a Wireshark packet capture window. The left pane shows a list of packets with timestamps. The middle pane shows the 'Statistics' tab, which is divided into 'Interfaces' and 'Statistics' sections. The 'Statistics' section is further divided into 'Measurement' and 'Captured' sub-sections. The 'Measurement' section shows various statistics for the captured data, including packets, time span, average pps, average packet size, bytes, average bytes/s, and average bits/s. The 'Captured' section shows the number of packets captured and displayed, along with the packet size limit. The right pane shows a list of packets with their details and raw data.

Interfaces	
Interface	Dropped packets
Unknown	Unknown

Statistics			
Measurement	Captured	Displayed	Marked
Packets	7824	219 (2.8%)	—
Time span, s	83.716	40.337	—
Average pps	93.5	5.4	—
Average packet size, B	2778	885	—
Bytes	21734943	193725 (0.9%)	0
Average bytes/s	259 k	4,802	—
Average bits/s	2,077 k	38 k	—

The longest time to load in the entire webpage was the html page because of the DNS resolution, TCP/TLS handshake, and server processing time. The assets load faster because they are being loaded in parallel for different contents and don't block the loading of the entire html page.

PART 2 (DNS)

QUES 1.

COMMAND USED : dns

Details

File

Name:

/home/ashish/final.pcap

Length:

21 MB

Hash (SHA256):

6dcd03523773da859203b2225eeea2406af469f029f5991d3580fe7535f9f8d5

Hash (SHA1):

3f86a9ae40e57e3b4ab6f2e0a9257145ad1301c1

Format:

Wireshark/tcpdump/... - pcap

Encapsulation:

Ethernet

Snapshot length:

262144

Time

First packet:

2026-02-07 16:37:47

Last packet:

2026-02-07 16:39:11

Elapsed:

00:01:23

Capture

Hardware:

Unknown

OS:

Unknown

Application:

Unknown

Interfaces

Interface

Unknown

Dropped packets

Unknown

Capture filter

Unknown

Link type

Ethernet

Packet size limit (snaplen)

262144 bytes

Statistics

Measurement

Packets

Time span, s

Average pps

Average packet size, B

Bytes

Average bytes/s

Average bits/s

Captured

7824

83.716

93.5

2778

21734943

259 k

2,077 k

Displayed

156 (2.0%)

60.030

2.6

197

30666 (0.1%)

510

4,086

Marked

2 (0.0%)

25.315

0.1

434

867 (0.0%)

34

273

Total of 156 DNS packets which includes for the iitbhilai.com , [netflix.com](#) we visited as well as google background processes, ads and various CDNs

a.)

Domain Name, IP Table :

Domain Name	IP Address
iitbhilai server	192.168.10.115
netflix.com	N/A
apiproxy-website-nlb-prod-2-e98cb8cf33ff3581.elb.us-west-2.amazonaws.com	44.234.232.238

apiproxy-website-nlb-prod-2-e98cb8cf33ff3581.elb.us-west-2.amazonaws.com	44.237.234.25
apiproxy-website-nlb-prod-2-e98cb8cf33ff3581.elb.us-west-2.amazonaws.com	44.242.60.85

netflix.com is alias of dradis.netflix.com which is an alias of other IP which is resolved for the aws ultimately. The next DNS request is resolved with multiple IP addresses of AWS. We have made the table for the queries we had made, for the remaining DNS queries they might be the browser or the google or website background process.

b.) The IP of the DNS server is 192.168.10.87 which might be our local DNS resolver handling the requests.

Wide ☐ Case sensitive String	
Source	Destination
10.50.33.117	192.168.10.87
10.50.33.117	192.168.10.87
192.168.10.87	10.50.33.117
192.168.10.87	10.50.33.117
10.50.33.117	192.168.10.87
10.50.33.117	192.168.10.87
10.50.33.117	192.168.10.87
10.50.33.117	192.168.10.87
192.168.10.87	10.50.33.117
192.168.10.87	10.50.33.117
192.168.10.87	10.50.33.117
192.168.10.87	10.50.33.117

QUES 2.

COMMAND USED : dig root-servers.net NS

```
ashish@ashish-ASUS-TUF-Gaming-F15-FX507VV-FX507VV:~/Desktop/CN/12340370$ dig root-servers.net NS

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> root-servers.net NS
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 48233
;; flags: qr rd ra; QUERY: 1, ANSWER: 13, AUTHORITY: 0, ADDITIONAL: 27

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;root-servers.net.                IN      NS

;; ANSWER SECTION:
root-servers.net. 155844 IN      NS      l.root-servers.net.
root-servers.net. 155844 IN      NS      i.root-servers.net.
root-servers.net. 155844 IN      NS      j.root-servers.net.
root-servers.net. 155844 IN      NS      k.root-servers.net.
root-servers.net. 155844 IN      NS      m.root-servers.net.
root-servers.net. 155844 IN      NS      c.root-servers.net.
root-servers.net. 155844 IN      NS      d.root-servers.net.
root-servers.net. 155844 IN      NS      b.root-servers.net.
root-servers.net. 155844 IN      NS      g.root-servers.net.
root-servers.net. 155844 IN      NS      h.root-servers.net.
root-servers.net. 155844 IN      NS      e.root-servers.net.
root-servers.net. 155844 IN      NS      f.root-servers.net.
root-servers.net. 155844 IN      NS      a.root-servers.net.

;; ADDITIONAL SECTION:
a.root-servers.net. 590982 IN      A       198.41.0.4
a.root-servers.net. 224573 IN      AAAA    2001:503:ba3e::2:30
b.root-servers.net. 587845 IN      A       170.247.170.2
b.root-servers.net. 587845 IN      AAAA    2801:1b8:10::b
c.root-servers.net. 155844 IN      A       192.33.4.12
c.root-servers.net. 155844 IN      AAAA    2001:500:2::c
d.root-servers.net. 587847 IN      A       199.7.91.13
d.root-servers.net. 587847 IN      AAAA    2001:500:2d::d
e.root-servers.net. 587845 IN      A       192.203.230.10
e.root-servers.net. 587845 IN      AAAA    2001:500:a8::e
f.root-servers.net. 587846 IN      A       192.5.5.241
f.root-servers.net. 587846 IN      AAAA    2001:500:2f::f
g.root-servers.net. 587847 IN      A       192.112.36.4
g.root-servers.net. 587847 IN      AAAA    2001:500:12::d0d
h.root-servers.net. 587846 IN      A       198.97.190.53
h.root-servers.net. 587846 IN      AAAA    2001:500:1::53
```

```
ashish@ashish-ASUS-TUF-Gaming-F15-FX507VV-FX507VV:~/Desktop/CN/12340370$ dig @198.41.0.4 www.iitbhilai.ac.in +norecurse

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> @198.41.0.4 www.iitbhilai.ac.in +norecurse
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 52928
;; flags: qr; QUERY: 1, ANSWER: 0, AUTHORITY: 4, ADDITIONAL: 9

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;www.iitbhilai.ac.in.            IN      A

;; AUTHORITY SECTION:
in. 172800 IN      NS      ns10.trs-dns.org.
in. 172800 IN      NS      ns01.trs-dns.com.
in. 172800 IN      NS      ns10.trs-dns.info.
in. 172800 IN      NS      ns01.trs-dns.net.

;; ADDITIONAL SECTION:
ns10.trs-dns.org. 172800 IN      A       64.78.205.1
ns10.trs-dns.org. 172800 IN      AAAA    2620:171:813:1534:8::1
ns01.trs-dns.com. 172800 IN      A       64.96.1.1
ns01.trs-dns.com. 172800 IN      AAAA    2620:57:4001::1
ns10.trs-dns.info. 172800 IN      A       64.78.204.1
ns10.trs-dns.info. 172800 IN      AAAA    2620:171:812:1534:8::1
ns01.trs-dns.net. 172800 IN      A       64.96.2.1
ns01.trs-dns.net. 172800 IN      AAAA    2620:57:4002::1

;; Query time: 243 msec
;; SERVER: 198.41.0.4#53(198.41.0.4) (UDP)
;; WHEN: Sat Feb 07 21:33:10 IST 2026
;; MSG SIZE rcvd: 345
```

```

See 'snap info <snapname>' for additional versions.
ashish@ashish-ASUS-TUF-Gaming-F15-FX507VY-FX507VY: /Desktop/CN12345678$ dig @dns1.iitbhilai.ac.in www.iitbhilai.ac.in +norecurse
; <<<> DIG 9.18.39-@ubuntu24.04.2-Ubuntu <<<> @dns1.iitbhilai.ac.in www.iitbhilai.ac.in +norecurse
; (1 server found)
; global options: +cmd
; Got answer:
; ->HEADER<- opcode: QUERY, status: NOERROR, id: 35473
; Flags: qr aa ra; QUERY: 1, ANSWER: 1, AUTHORITY: 1, ADDITIONAL: 2
; OPT PSEUDOSECTION:
; EDNS: version: 0, flags: 0, udp: 4096
; QUESTION SECTION:
; www.iitbhilai.ac.in.      IN      A
; ANSWER SECTION:
; www.iitbhilai.ac.in.      641     IN      A      192.168.10.115
; AUTHORITY SECTION:
; iitbhilai.ac.in.         641     IN      NS      dns2.iitbhilai.ac.in.
; ADDITIONAL SECTION:
; dns2.iitbhilai.ac.in.    641     IN      A      192.168.10.72
; Query time: 3 msec
; SERVER: 192.168.10.87#53(dns1.iitbhilai.ac.in) (UDP)
; WHEN: Sat Feb 07 21:35:30 IST 2026
; MSG SIZE  rcvd: 99
ashish@ashish-ASUS-TUF-Gaming-F15-FX507VY-FX507VY: /Desktop/CN12345678$

; Query time: 243 msec
; SERVER: 198.41.0.4#53(198.41.0.4) (UDP)
; WHEN: Sat Feb 07 21:33:10 IST 2026
; MSG SIZE  rcvd: 345
ashish@ashish-ASUS-TUF-Gaming-F15-FX507VY-FX507VY: /Desktop/CN12345678$ dig @ns10.trs-dns.org www.iitbhilai.ac.in +norecurse
; <<<> DIG 9.18.39-@ubuntu24.04.2-Ubuntu <<<> @ns10.trs-dns.org www.iitbhilai.ac.in +norecurse
; (2 servers found)
; global options: +cmd
; Got answer:
; ->HEADER<- opcode: QUERY, status: NOERROR, id: 44634
; Flags: qr; QUERY: 1, ANSWER: 0, AUTHORITY: 2, ADDITIONAL: 3
; OPT PSEUDOSECTION:
; EDNS: version: 0, flags: 0, udp: 1232
; QUESTION SECTION:
; www.iitbhilai.ac.in.      IN      A
; AUTHORITY SECTION:
; iitbhilai.ac.in.         900     IN      NS      dns1.iitbhilai.ac.in.
; iitbhilai.ac.in.         900     IN      NS      dns2.iitbhilai.ac.in.
; ADDITIONAL SECTION:
; dns1.iitbhilai.ac.in.    900     IN      A      103.147.138.110
; dns2.iitbhilai.ac.in.    900     IN      A      103.147.138.111
; Query time: 93 msec
; SERVER: 64.74.205.145(ns10.trs-dns.org) (UDP)
; WHEN: Sat Feb 07 21:33:55 IST 2026
; MSG SIZE  rcvd: 118
ashish@ashish-ASUS-TUF-Gaming-F15-FX507VY-FX507VY: /Desktop/CN12345678$

```

The final resolved IP is 192.168.10.115 as seen from the Answer section.

List of servers involved :

root-servers.net

198.41.0.4

ns10.trs-dns.org

dns1.iitbhilai.ac.in